

Practical 5

Aim:

To install MongoDB and perform CRUD operations and data aggregation using MongoDB.

Problem definition:

Perform Create, Read, Update, and Delete (CRUD) operations using MongoDB. Students will work with a NoSQL document database to understand how unstructured and semi structured data is managed at scale in Big Data environments.

Objectives:

Tasks:

1. Install and configure MongoDB.
2. Create a database and collection.
3. Insert documents (single and multiple).
4. Read/Query documents using various filters.
5. Update existing documents.
6. Delete documents and collections.
7. Use MongoDB aggregation pipeline for data analysis.

Observation / Output :

1. Install and configure MongoDB.

```
Shalvi_02@DESKTOP-9UVK663:~$ sudo apt update
[sudo] password for Shalvi_02:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1263 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [292 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1807 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1686 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [337 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [213 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1550 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1207 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [355 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [242 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.3 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1445 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [336 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [45.7 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [12.8 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:24 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5772 B]
Get:25 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Fetched 11.1 MB in 13s (837 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
109 packages can be upgraded. Run 'apt list --upgradable' to see them.
Shalvi_02@DESKTOP-9UVK663:~$ sudo apt install mongodb -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package mongodb is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source
E: Package 'mongodb' has no installation candidate
```

24DIT050

CSUE301 : Big Data Analytics

```
mongosh mongodb://127.0.0.1:27021
E: Package 'mongodb' has no installation candidate
Shalvi_02@DESKTOP-9UVK663:~$ apt-cache policy mongodb
mongodb:
  Installed: (none)
  Candidate: (none)
  Version table:
Shalvi_02@DESKTOP-9UVK663:~$ E: Package 'mongodb' has no installation candidate
Shalvi_02@DESKTOP-9UVK663:~$ apt-cache policy mongodb
mongodb:
  Installed: (none)
  Candidate: (none)
  Version table:
Shalvi_02@DESKTOP-9UVK663:~$
E: command not found
Shalvi_02@DESKTOP-9UVK663:~$ command not found
mongodb: command not found
-bash: syntax error near unexpected token `none'
-bash: syntax error near unexpected token `none'
Version: command not found
Shalvi_02@DESKTOP-9UVK663:~$ command not found
Shalvi_02@DESKTOP-9UVK663:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description: Ubuntu 24.04.4 LTS
Release: 24.04
Codename: noble
Shalvi_02@DESKTOP-9UVK663:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description: Ubuntu 24.04.4 LTS
Release: 24.04
Codename: noble
Shalvi_02@DESKTOP-9UVK663:~$ sudo apt-get update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Shalvi_02@DESKTOP-9UVK663:~$ sudo apt-get install -y gnupg curl
Reading package lists... Done
Building dependency tree... Done
```

```
Reading package lists... Done
Shalvi_02@DESKTOP-9UVK663:~$ sudo apt-get install -y gnupg curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amd-gpu libglapi-mesa libllvm19
  libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  dirnmgpr gnupg-l10n gnupg-utils gpg gpg-agent gpg-wks-client gpgconf
  gpgsm gpgv keyboxd libcurl3t64-gnutls libcurl4t64
Suggested packages:
  pinentry-gnome3 tor parcimonie xloadimage gpg-wks-server scdaemon
The following packages will be upgraded:
  curl dirnmgpr gnupg gnupg-l10n gnupg-utils gpg gpg-agent gpg-wks-client
  gpgconf gpgsm gpgv keyboxd libcurl3t64-gnutls libcurl4t64
14 upgraded, 0 newly installed, 0 to remove and 95 not upgraded.
Need to get 2293 kB/3197 kB of archives.
After this operation, 3072 B of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gpg-wks-client amd64 2.4.4-2ubuntu17.6 [70.9 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 dirnmgpr amd64 2.4.4-2ubuntu17.6 [323 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gpgsm amd64 2.4.4-2ubuntu17.6 [232 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gnupg-utils amd64 2.4.4-2ubuntu17.6 [109 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gpg-agent amd64 2.4.4-2ubuntu17.6 [227 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gpg amd64 2.4.4-2ubuntu17.6 [565 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gpgconf amd64 2.4.4-2ubuntu17.6 [104 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gnupg all 2.4.4-2ubuntu17.6 [359 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 keyboxd amd64 2.4.4-2ubuntu17.6 [78.3 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gpgv amd64 2.4.4-2ubuntu17.6 [158 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gnupg-l10n all 2.4.4-2ubuntu17.6 [66.5 kB]
Fetched 2293 kB in 7s (344 kB/s)
(Reading database ... 69526 files and directories currently installed.)
Preparing to unpack .../0-gpg-wks-client_2.4.4-2ubuntu17.6_amd64.deb ...
Unpacking gpg-wks-client (2.4.4-2ubuntu17.6) over (2.4.4-2ubuntu17.4) ...
Preparing to unpack .../1-dirnmgpr_2.4.4-2ubuntu17.6_amd64.deb ...
Unpacking dirnmgpr (2.4.4-2ubuntu17.6) over (2.4.4-2ubuntu17.4) ...
Preparing to unpack .../2-gpgsm_2.4.4-2ubuntu17.6_amd64.deb ...
Unpacking gpgsm (2.4.4-2ubuntu17.6) over (2.4.4-2ubuntu17.4) ...
Preparing to unpack .../3-gnupg-utils_2.4.4-2ubuntu17.6_amd64.deb ...
Unpacking gnupg-utils (2.4.4-2ubuntu17.6) over (2.4.4-2ubuntu17.4) ...
```

24DIT050

CSUE301 : Big Data Analytics

```
mongosh mongodb://127.0.0.1 x + v
--dearmor
Shalvi_02@DESKTOP-9UVK663:~$ echo "deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-8.0.gpg ] https://repo.mongodb.org/apt/ubuntu noble/m
ongodb-org/8.0 multiverse" | sudo tee /etc/apt/sources.list.d/mongodb-org-8.0.list
Shalvi_02@DESKTOP-9UVK663:~$ echo "deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-8.0.gpg ] https://repo.mongodb.org/apt/ubuntu noble/m
ongodb-org/8.0 multiverse" | sudo tee /etc/apt/sources.list.d/mongodb-org-8.0.list
Shalvi_02@DESKTOP-9UVK663:~$ echo "deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-8.0.gpg ] https://repo.mongodb.org/apt/ubuntu noble/m
ongodb-org/8.0 multiverse" | sudo tee /etc/apt/sources.list.d/mongodb-org-8.0.list
Shalvi_02@DESKTOP-9UVK663:~$ sudo apt-get update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Get:3 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0 InRelease [3005 B]
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:6 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0/multiverse amd64 Packages [89.8 kB]
Get:7 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0/multiverse arm64 Packages [89.5 kB]
Fetched 182 kB in 8s (23.6 kB/s)
Reading package lists... Done
Shalvi_02@DESKTOP-9UVK663:~$ sudo apt install mongodb
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package mongodb is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

E: Package 'mongodb' has no installation candidate
Shalvi_02@DESKTOP-9UVK663:~$ sudo apt-get install -y mongodb-org
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amd-glx libglapi-mesa libllvm19
  libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  mongodb-database-tools mongodb-mongosh mongodb-org-database
  mongodb-org-database-tools-extra mongodb-org-mongos mongodb-org-server
  mongodb-org-shell mongodb-org-tools
0 upgraded, 10 newly installed, 0 to remove and 0 not upgraded.
Need to get 182 kB of archives.
After this operation, 412 kB of additional disk space will be used.
Get:1 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0/multiverse amd64 mongodb-org-shell 8.0.1 [182 kB]
Fetched 182 kB in 8s (23.6 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package mongodb-org-shell.
(Reading database ... 123456 files and directories currently installed.)
Preparing to unpack .../mongodb-org-shell_8.0.1_amd64.deb ...
Unpacking mongodb-org-shell (8.0.1) ...
Setting up mongodb-org-shell (8.0.1) ...
```

2. Create a database and collection.

```
mongosh mongodb://127.0.0.1 x + v
L-2556 /usr/bin/mongod --config /etc/mongod.conf
Sep 15 12:38:26 DESKTOP-9UVK663 systemd[1]: Started mongod.service - MongoDB>
Sep 15 12:38:26 DESKTOP-9UVK663 mongod[2556]: {'t': {'$date': '2026-09-15T12:38:26.000Z'}}
lines=12/12 (END)
Shalvi_02@DESKTOP-9UVK663:~$ mongosh
Current Mongosh Log ID: 6aa93be0187f959ce0db2533
Connecting to:
  mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.11.1
Using MongoDB:
  8.0.32
Using Mongosh:
  2.11.1

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

To help improve our products, pseudo-anonymous usage data is collected from interactive and agentic sessions only, and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

-----
The server generated these startup warnings when booting
2026-09-15T12:38:26.977+00:00: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/pr
odnotes-filesystem
2026-09-15T12:38:27.237+00:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2026-09-15T12:38:27.338+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2026-09-15T12:38:27.338+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2026-09-15T12:38:27.338+00:00: We suggest setting the contents of sysfsFile to 0.
2026-09-15T12:38:27.338+00:00: We suggest setting swappiness to 0 or 1, as swapping can cause performance problems.
-----

test> show dbs
admin    40.00 KiB
config  12.00 KiB
local    40.00 KiB
test> use college
switched to db college
college>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
college> db.createCollection("students")
{ ok: 1 }
college>
```

```
test> show dbs
admin    40.00 KiB
config  12.00 KiB
local    40.00 KiB
test> use college
switched to db college
college>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
college> db.createCollection("students")
{ ok: 1 }
college> show collections
students
college> |
```

24DIT050

CSUE301 : Big Data Analytics

3. Insert documents (single and multiple).

```
test> show dbs
admin    48.00 KiB
config  12.00 KiB
local    48.00 KiB
test> use college
switched to db college
college>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
college> db.createCollection("students")
{ ok: 1 }
college> show collections
students
college> db.students.insertOne({
  name: "Shalvi",
  age: 21,
  course: "IT",
  semester: 5,
  marks: 85
})
{
  acknowledged: true,
  insertedId: ObjectId('6aa93db1187f959ce0db2534')
}
college> db.students.insertOne({
  name: "Shalvi",
  age: 21,
  course: "IT",
  semester: 5,
  marks: 85
})
{
  acknowledged: true,
  insertedId: ObjectId('6aa93dbe187f959ce0db2535')
}
college>
```

4. Read/Query documents using various filters.

```
mongosh mongod://127.0.0.1:27020
students
college> db.students.insertOne({
  name: "Shalvi",
  age: 21,
  course: "IT",
  semester: 5,
  marks: 85
})
{
  acknowledged: true,
  insertedId: ObjectId('6aa93db1187f959ce0db2534')
}
college> db.students.insertOne({
  name: "Shalvi",
  age: 21,
  course: "IT",
  semester: 5,
  marks: 85
})
{
  acknowledged: true,
  insertedId: ObjectId('6aa93dbe187f959ce0db2535')
}
college> db.students.find()
[
  {
    _id: ObjectId('6aa93db1187f959ce0db2534'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  },
  {
    _id: ObjectId('6aa93dbe187f959ce0db2535'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  }
]
```

24DIT050

CSUE301 : Big Data Analytics

```
mongosh mongodb://127.0.0.1:27020
> use college
> db.students.findOne({name: "Shalvi"})
{
  _id: ObjectId('6aa93db1187f959ce0db2534'),
  name: 'Shalvi',
  age: 21,
  course: 'IT',
  semester: 5,
  marks: 85
}
> db.students.find({course: "IT"})
[
  {
    _id: ObjectId('6aa93db1187f959ce0db2534'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  },
  {
    _id: ObjectId('6aa93dbe187f959ce0db2535'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  }
]
> db.students.find({marks: {$gt: 80}})
[
  {
    _id: ObjectId('6aa93db1187f959ce0db2534'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  },
  {
    _id: ObjectId('6aa93dbe187f959ce0db2535'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  }
]
```

```
mongosh mongodb://127.0.0.1:27020
> use college
> db.students.find({marks: {$gte: 80}})
[
  {
    _id: ObjectId('6aa93db1187f959ce0db2534'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  },
  {
    _id: ObjectId('6aa93dbe187f959ce0db2535'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  }
]
> db.students.find({marks: {$lt: 80}})
[]
> db.students.find({marks: {$gte: 80, $lte: 90}})
[
  {
    _id: ObjectId('6aa93db1187f959ce0db2534'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  },
  {
    _id: ObjectId('6aa93dbe187f959ce0db2535'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  }
]
> db.students.find({course: "IT", marks: {$gt: 80}})
[
  {
    _id: ObjectId('6aa93db1187f959ce0db2534'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  },
  {
    _id: ObjectId('6aa93dbe187f959ce0db2535'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  }
]
```

```
mongosh mongodb://127.0.0.1:27020
> use college
> db.students.find({course: "IT", marks: {$gt: 80}})
[
  {
    _id: ObjectId('6aa93db1187f959ce0db2534'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  },
  {
    _id: ObjectId('6aa93dbe187f959ce0db2535'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  }
]
>
```

```
college> db.students.find(
  {
    $or: [
      {course: "IT"},
      {course: "CSE"}
    ]
  })
[
  {
    _id: ObjectId('6aa93db1187f959ce0db2534'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  },
  {
    _id: ObjectId('6aa93dbe187f959ce0db2535'),
    name: 'Shalvi',
    age: 21,
    course: 'IT',
    semester: 5,
    marks: 85
  }
]
college> db.students.find(
  {
    {name: 1, marks: 1, _id: 0}
  }
)
[ { name: 'Shalvi', marks: 85 }, { name: 'Shalvi', marks: 85 } ]
college> |
```

5. Update existing documents.

```
mongosh mongodb://127.0.0.1:27020
[ { name: 'Shalvi', marks: 85 }, { name: 'Shalvi', marks: 85 } ]
college> db.students.updateOne(
  {
    {name: "Shalvi"},
    {$set: {marks: 90}}
  }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
college> db.students.findOne({name: "Shalvi"})
{
  _id: ObjectId('6aa93db1187f959ce0db2534'),
  name: 'Shalvi',
  age: 21,
  course: 'IT',
  semester: 5,
  marks: 90
}
college> db.students.updateMany(
  {
    {course: "IT"},
    {$set: {semester: 6}}
  }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
college> db.students.updateMany(
  {
    {course: "IT"},
    {$inc: {marks: 5}}
  }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
college> |
```

6. Delete documents and collections.

```
college>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
college> db.students.deleteOne({
  |   name: "Amit"
  | })
{ acknowledged: true, deletedCount: 0 }
college> db.students.deleteMany({
  |   marks: {$lt: 80}
  | })
{ acknowledged: true, deletedCount: 0 }
college> db.students.deleteMany({})
{ acknowledged: true, deletedCount: 2 }
college> db.students.find()

college> db.students.drop()
true
college> show collections

college> |
```

7. Use MongoDB aggregation pipeline for data analysis.

```
mongosh mongodb://127.0.0.1:27000
college> db.students.insertMany([
  {name: "Shalvi", course: "IT", marks: 90},
  {name: "Rahul", course: "CSE", marks: 70},
  {name: "Neha", course: "CE", marks: 92},
  {name: "Amit", course: "IT", marks: 88},
  {name: "Priya", course: "CSE", marks: 70},
  {name: "Riya", course: "IT", marks: 82}
])
acknowledged: true,
insertedIds: [
  ObjectId('6aa941f2187f959ce0db2536'),
  ObjectId('6aa941f2187f959ce0db2537'),
  ObjectId('6aa941f2187f959ce0db2538'),
  ObjectId('6aa941f2187f959ce0db2539'),
  ObjectId('6aa941f2187f959ce0db253a'),
  ObjectId('6aa941f2187f959ce0db253b')
]
college> db.students.aggregate([
  {
    $group: {
      _id: null,
      averageMarks: {$avg: "$marks"}
    }
  }
])
[ { _id: null, averageMarks: 84.16666666666667 } ]
college> db.students.aggregate([
  {
    $group: {
      _id: "$course",
      averageMarks: {$avg: "$marks"}
    }
  }
])
[ { _id: 'IT', averageMarks: 86.66666666666667 },
  { _id: 'CSE', averageMarks: 70.5 },
  { _id: 'CE', averageMarks: 92 } ]
college> db.students.aggregate([
  {
    $group: {
      _id: null,
      highestMarks: {$max: "$marks"}
    }
  }
])
[ { _id: null, highestMarks: 92 } ]
college> db.students.aggregate([
  {
    $group: {
      _id: null,
      lowestMarks: {$min: "$marks"}
    }
  }
])
```

```
college> db.students.aggregate([
  {
    $group: {
      _id: "$course",
      totalStudents: {$sum: 1}
    }
  }
])
[ { _id: 'IT', totalStudents: 3 },
  { _id: 'CSE', totalStudents: 2 },
  { _id: 'CE', totalStudents: 1 } ]
college> db.students.aggregate([
  {
    $sort: {marks: -1}
  }
])
[ { _id: ObjectId('6aa941f2187f959ce0db2538'),
  name: 'Neha',
  course: 'CE',
  marks: 92 },
  { _id: ObjectId('6aa941f2187f959ce0db2536'),
  name: 'Unnati',
  course: 'IT',
  marks: 90 },
  { _id: ObjectId('6aa941f2187f959ce0db2539'),
  name: 'Amit',
  course: 'IT',
  marks: 88 },
  { _id: ObjectId('6aa941f2187f959ce0db253b'),
  name: 'Riya',
  course: 'IT',
  marks: 82 } ]
```

```
college> db.students.aggregate([
  {
    $match: {course: "IT"}
  },
  {
    $sort: {marks: -1}
  }
])
[ { _id: ObjectId('6aa941f2187f959ce0db2536'),
  name: 'Shalvi',
  course: 'IT',
  marks: 90 },
  { _id: ObjectId('6aa941f2187f959ce0db2539'),
  name: 'Amit',
  course: 'IT',
  marks: 88 },
  { _id: ObjectId('6aa941f2187f959ce0db253b'),
  name: 'Riya',
  course: 'IT',
  marks: 82 } ]
college>
```

Result :

MongoDB was successfully installed and configured. CRUD operations and aggregation were performed successfully on the MongoDB database.

Key Questions /Analysis:

Q1. What is MongoDB and how does it differ from relational databases?

MongoDB is a NoSQL document database that stores data in flexible documents. Unlike relational databases, it uses collections and documents instead of tables and rows.

Q2. What are CRUD operations and why are they fundamental?

CRUD means Create, Read, Update, Delete. They are fundamental because they allow users to add, view, modify, and remove data.

Q3. How does MongoDB store data (BSON format)?

MongoDB stores data as BSON (Binary JSON) documents. BSON supports different data types such as strings, numbers, arrays, dates, and objects.

Q4. What is the aggregation pipeline in MongoDB?

The aggregation pipeline is a sequence of stages used to process and analyze data, such as filtering, grouping, sorting, and calculating averages.

Q5. When should NoSQL databases be preferred over SQL databases?

NoSQL should be preferred for large, unstructured/semi-structured, rapidly changing, and distributed data where flexible schema and scalability are important.

Conclusion:

In this practical, MongoDB was installed and configured in Ubuntu. A database and collection were created, and single and multiple documents were inserted. CRUD operations were performed using insert, find, update, and delete commands. Finally, the MongoDB aggregation pipeline was used to perform data analysis such as average, maximum, minimum, counting, grouping, and sorting.